# Anfrage Kabelablängen
## Fa. Metzner

Modulare Automation

Gluth SYSTEM TECHNIK

# Aufgabenstellung



Herstellen einer Anlage zur automatischen Konfektionierung und Prüfung von Kabeln.

## Auszüge aus dem Lastenheft:

| Designation | Amount | Material number |
|---|---|---|
| Cable (Low Voltage singlecore cable 16mm²/25mm² /35mm² FLRY-B;FLR2G;)  | 1 pc | L0263CXX (16mm²)<br>L3363066 (16mm²)<br>L3366063 (25mm²)<br>L3369048 (35mm²) |

| | |
|---|---|
| Dimensions [mm] (Cable Length) | 300 - 1500 |

Geändert: 170 … 1500 mm

### 15.4 Cutting to length

1. Cable entry
   - Reliably inputting of the cable, cable transport mustn´t be hindered by machine parts
   - Conical straightening device, declining with at least 4 + 5 rolls, scale and quick-fastener
   - Cable feeding: min 2,5m / sec, freely programmable
   - Acceleration freely programmable

   - Length repeat accuracy:
     - Cable length smaller 500mm < +/-1mm
     - Cable length >=500mm < 0,2%
   - Observing the + tolerance area has to be adjustable by correction factor
   - Mechanical knot control ( diameter specific and marked adjusting disc)
   - Minimum cable waste ( 1. Cable, zero cut)

2. Cut-in units
   - Pre-switching of a cable roller with cable store, types or power specified by the consultant
   - Interface description cut-in units
   - Automatic identification if the actual stripping blade is the correct one for the loaded order.

3. Length cutting
   - Length area: 500 - 2000mm
   - Cutting blades
     - Blade quality: standard materials and options, tolerances
     - Lifetime for each type

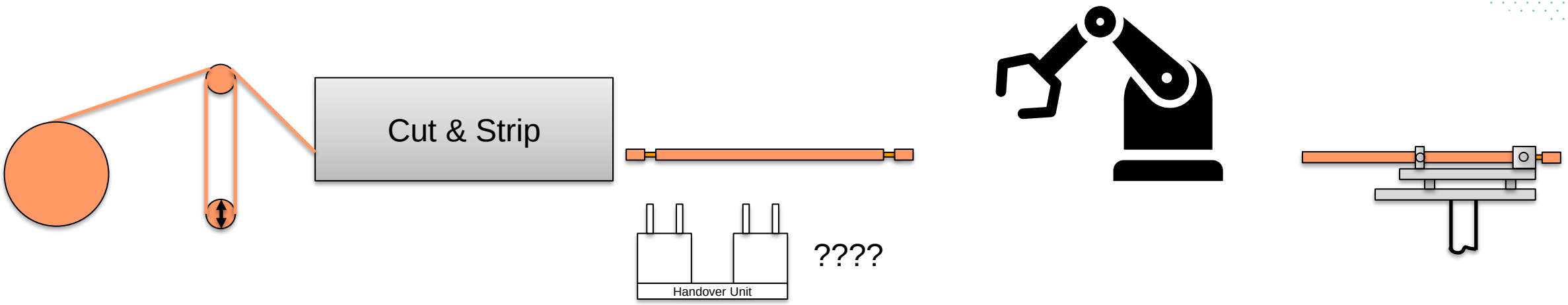
Geändert: 170 … 1500 mm

### 15.5 Stripping

1. Stripping length of each individual line 2 - 20mm, tolerance +/- 0.2mm;
   - Freely adjustable
   - Adjustable in 0,1mm steps
2. The stripping length is transferred via MES including tolerance.
3. Knifes ( specified from the supplier )
   - Stripping knifes
   - Gradation based on the wire dimensions
   - Types: V-blades, radius-V-blades, radius-blades
4. Blade quality: standard material and options
5. Lifetime per model
6. Integrated vacuum cleaning
   - No clogging of the suction hoses
   - Collection container must be designed in such a way that the container only has to be emptied once per shift (8 hours)
   - Collection container with quick change system
   - 2 collection containers should be offered

# Konzept Kabelzuführung






| Feeding Unit | Cut & Strip Unit | Handover Unit | Robot Handling | Round Table |
|---|---|---|---|---|

Lieferant ???? Gluth

# Anlieferung Kabel



## Anlieferung der Kabel:

| Cable type / Leitungstyp | | | | | | Standard Reel / Standartspule | |
|---|---|---|---|---|---|---|---|
| Order No. Bestellnr. | | Type Typ | Cross-section Querschnitt | Stranding Acc. to ISO 19642-3 (tab A2) | Weight Gewicht approx./ca. | Make-Up Aufmachung | Content Füllmenge approx./ca. |
| Bare/Blank | Tinned/Verzinnt | | [mm²] | | [g/m] | | [m/Reel] |
| LRYA1600xxyy | LRYSA1600xxyy | FLRY-A | 16,00 | - | 158 | D 600 | 1 000 |
| LRY1600xxyy | LRYSN1600xxyy | FLRY-B | 16,00 | - | 158 | D 600 | 1 000 |
| LRYS1600xxyy | LRYSS1600xxyy | FLRY-S | 16,00 | Standard | 158 | D 600 | 1 000 |
| LRYF1600xxyy | LRYSF1600xxyy | FLRY-C | 16,00 | Flexible | 158 | D 600 | 1 000 |
| LRYA2500xxyy | LRYSA2500xxyy | FLRY-A | 25,00 | - | 247 | D 800 | 1800 |
| LRY2500xxyy | LRYSN2500xxyy | FLRY-B | 25,00 | - | 247 | D 800 | 1800 |
| LRYS2500xxyy | LRYSS2500xxyy | FLRY-S | 25,00 | Standard | 247 | D 800 | 1800 |
| LRYF2500xxyy | LRYSF2500xxyy | FLRY-C | 25,00 | Flexible | 247 | D 800 | 1800 |
| LRY3000xxyy | LRYSN3000xxyy | FLRY-B | 30,00 | - | 289 | D 800 | 1400 |
| LRYS3000xxyy | LRYSS3000xxyy | FLRY-S | 30,00 | Standard | 289 | D 800 | 1400 |
| LRYF3000xxyy | LRYSF3000xxyy | FLRY-C | 30,00 | Flexible | 289 | D 800 | 1400 |
| LRY3500xxyy | LRYSN3500xxyy | FLRY-B | 35,00 | - | 351 | D 800 | 1100 |
| LRYS3500xxyy | LRYSS3500xxyy | FLRY-S | 35,00 | Standard | 351 | D 800 | 1100 |
| LRYF3500xxyy | LRYSF3500xxyy | FLRY-C | 35,00 | Flexible | 351 | D 800 | 1100 |
| VWYF3500xxyy | - | FLRY/0,16 | 35,00 | - | 351 | D 800 | 1100 |

ca. 160 kg

ca. 450 kg

ca. 385 kg

# Rahmendaten



Taktzeit: 6 Sekunden
Übergeordnete Steuerung: Siemens
Vorgabe der Kabelparameter durch die übergeordnete Steuerung

Teilabzug und Vollabzug beim Abisolieren geplant
„Handover Unit“ soll das Kabel übernehmen ~~und in U-Form bringen~~ (Klärung ob Lieferant oder Gluth), alternative: direkt von Roboter übernehmen

Roboter übernimmt das Kabel und übergibt es an das Transfersystem (Gluth)